

Pharmaceutical Sales Analysis for CEVO - BreatheEZ (Asthma Drug Relaunch)



Prepared by: Sakshi Kamble

Tools Used: MS Excel and PowerBI

Project Type: Pharmaceutical Case Study

Total Rows Analyzed: 2500

Problem Statement:-

CEVO Pharma launched a new asthma drug, *BreatheEZ*, in 2024 with the expectation of achieving strong market adoption in India. However, despite significant investments in marketing and sales efforts, the drug failed to generate the expected revenue and resulted in substantial losses.

The company now plans to relaunch the drug in 2026 and requires a data-driven analysis to understand the reasons behind the poor performance and to identify strategies for improving future sales outcomes.

Objective:-

The objective of this project is to analyze the 2024 pharmaceutical sales data to identify the key factors that led to the poor performance of the drug *BreatheEZ*. The analysis focuses on evaluating regional performance, doctor engagement, marketing effectiveness, and pricing strategy.

The project aims to answer the following business questions:

1. Why did the drug fail to achieve expected sales in 2024?
2. Which regions and cities should be targeted for the 2026 relaunch?
3. Which doctor specializations should be prioritized?
4. Should the pricing strategy be revised?
5. Which marketing channels should receive more investment?

Data Description:-

The dataset consists of **2,500 transactional records** from the year 2024, capturing detailed information about the sales and marketing performance of the drug *BreatheEZ*.

It includes variables related to geographic distribution, hospital-level sales, doctor specialization, marketing strategies, pricing, and revenue generation.

- **Date** - The date on which the drug sale transaction occurred
- **Company_Name** - Name of the pharmaceutical company (CEVO Pharma)
- **Drug_Name** - Name of the drug (*BreatheEZ*)
- **Region** - Geographic region where the sale occurred (North, South, East, West)
- **City** - City where the hospital is located
- **Hospital_ID** - Unique identifier for each hospital
- **Doctor_Specialization** - Type of doctor prescribing the drug (Pulmonologist, Pediatrician, etc.)
- **Sales_Rep_ID** - Unique identifier for each sales representative
- **Rep_Visits** - Number of visits made by sales representatives
- **Marketing_Channel** - Type of marketing strategy used (Webinar, Email, Medical Conference, Sales Visit)
- **Marketing_Spend** - Amount spent on marketing activities (in USD)
- **Price_per_unit** - Selling price of CEVO's drug per unit (in USD)
- **Competitor_Price** - Price of competing drugs in the market (in USD)
- **Units_Sold** - Number of units sold in the transaction
- **Revenue** - Total revenue generated from the sale (in USD)
- **Insurance_Coverage** - Indicates whether the drug purchase was covered by insurance (Yes/No)

As part of data preprocessing, additional features were created from the Date column, including **Month** and **Month_Name**, to enable time-based analysis and identify seasonal trends in drug sales.

Note: The dataset used in this analysis is a simulated dataset created for educational purposes. The company name (CEVO Pharma) and the drug name (BreatheEZ) are fictional and have been used to represent a realistic pharmaceutical business scenario. The dataset is designed to reflect real-world dynamics such as marketing strategies, pricing, and doctor engagement.

Methodology:-

The analysis was conducted using a structured approach to understand the factors affecting the performance of the drug *BreatheEZ* in 2024.

1. Data Understanding

The dataset was reviewed to understand key variables such as sales, marketing spend, pricing, doctor specialization, and regional distribution.

2. Data Preparation

Basic preprocessing was performed, including creating additional features such as Month and Month Name from the Date column to enable time-based analysis.

3. Exploratory Data Analysis (EDA)

Pivot tables and charts were created in Excel to analyze:

- Regional performance
- Doctor specialization contribution
- Marketing channel effectiveness
- Monthly sales trends

4. KPI and Metric Creation

Key performance indicators such as total revenue, total marketing spend, units sold, and ROI (Return on Investment) were calculated to evaluate business performance.

5. Data Visualization

An interactive dashboard was developed in Power BI to present insights using visualizations such as bar charts, line charts, and scatter plots.

6. Insight Generation

Patterns and trends identified from the analysis were used to derive key business insights and recommendations for the 2026 relaunch strategy.

Key Findings:-

- The company incurred a significant loss in 2024, with total revenue of approximately **\$1.13M** compared to a marketing spend of **\$2.81M**, resulting in a loss of around **\$1.69M**.
- CEVO's drug pricing was higher than competitors, with an average price of approximately **\$22.98 per unit** compared to the competitor's **\$19.98**, which likely impacted product adoption.
- Marketing channel analysis showed that **medical conferences had high spending but low returns**, whereas **webinars and sales visits delivered relatively better ROI (Return On Investment)**.

- Pulmonologists contributed the highest share of prescriptions, accounting for approximately **19.2K of total units sold**, while pediatricians contributed significantly less at around **9.5K**.
- Regional analysis revealed that the **East region generated the highest revenue (\$303K)**, while the **South region contributed the least (\$258K)**, indicating uneven market performance.
- Sales were concentrated among a limited number of hospitals, with the top hospitals contributing a significant portion of total units sold.
- A positive relationship was observed between sales representative visits and units sold, indicating that increased field engagement contributed to higher sales.

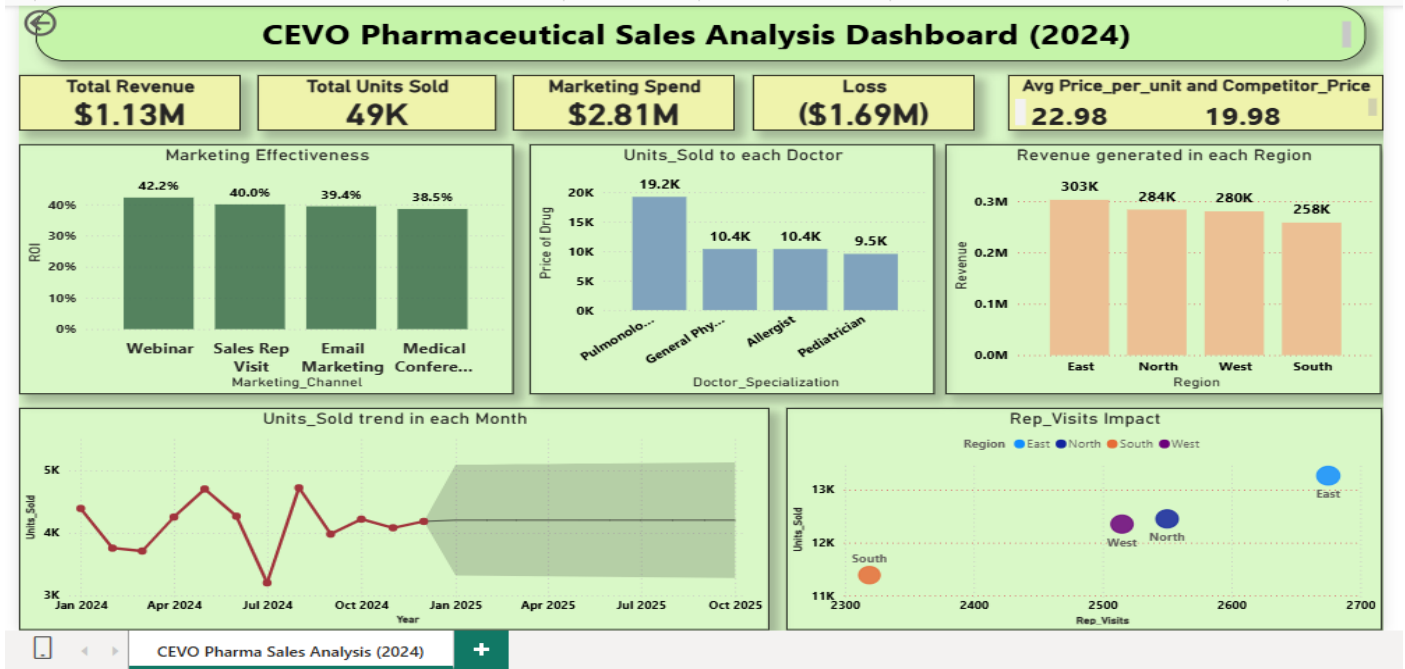


Fig 1: CEVO Pharmaceutical Sales Analysis Dashboard

Recommendations:-

1. CEVO should reconsider its pricing strategy, as the drug was priced higher (~\$22) than competitors (~\$19), which likely reduced adoption. A more competitive pricing approach or targeted discounts can help improve market penetration.
2. Marketing investments should be reallocated toward high-performing channels such as webinars and sales visits, which showed better ROI, while reducing spend on medical conferences, which delivered low returns.
3. CEVO should prioritize engagement with pulmonologists and allergists through targeted marketing and personalized outreach strategies.
4. A positive relationship between sales visits and units sold indicates that field engagement drives sales. CEVO should optimize visit frequency and ensure better coverage in high-potential areas.
5. Sales were concentrated among a limited number of hospitals. CEVO should strengthen relationships with top-performing hospitals and replicate successful strategies in similar healthcare institutions such as H1011, H1089 and H1039.
6. Sales trends indicate higher performance in certain months like August, May, January and June. CEVO should align marketing campaigns and product promotions with peak demand periods to maximize impact.

Conclusion:- The analysis of CEVO Pharma's 2024 drug launch revealed that high marketing spend, inefficient channel allocation, and higher pricing compared to competitors led to poor sales performance and overall losses.

However, the study also identified key opportunities in targeted doctor engagement, optimized marketing strategies, and regional focus. By implementing these data-driven recommendations, CEVO can improve market adoption and achieve a more successful and profitable relaunch in 2026.